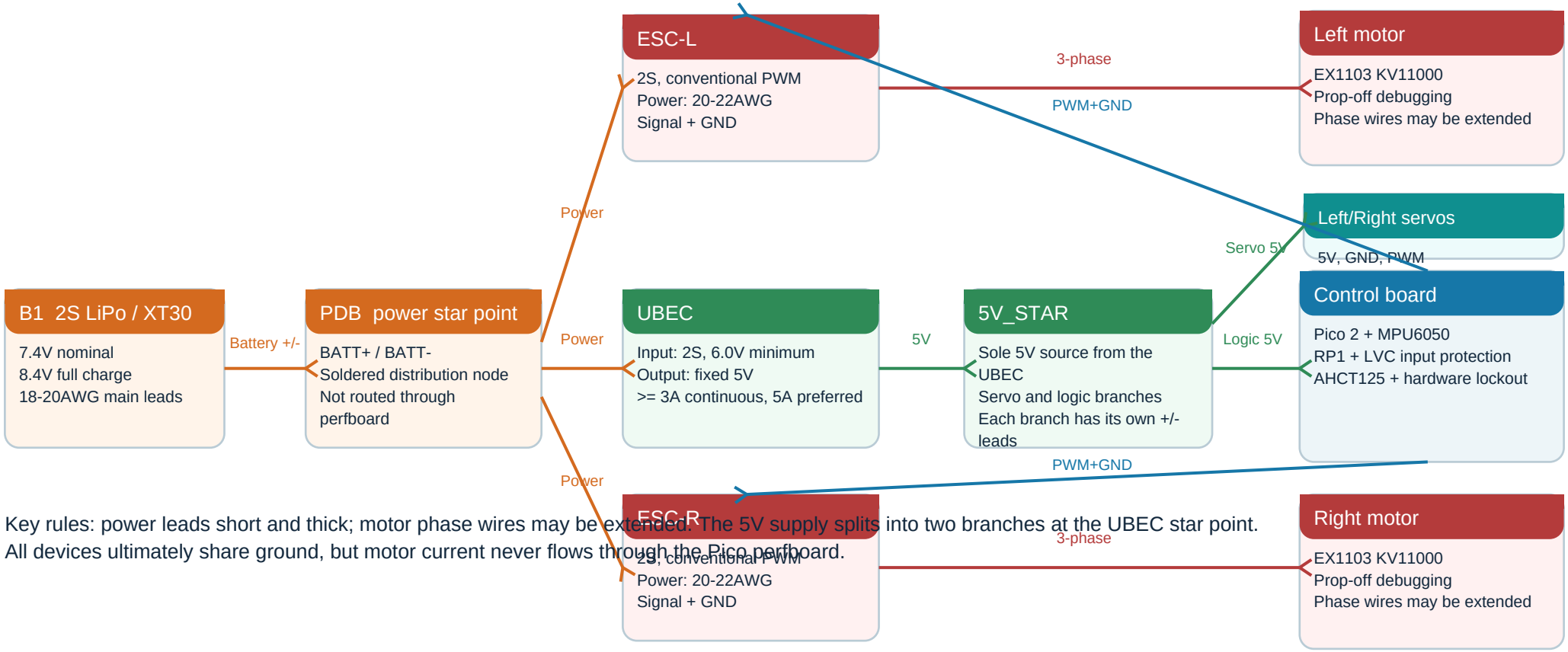
Overall power and signal architecture

Power current, 5V supply and control signals are routed as three separate paths



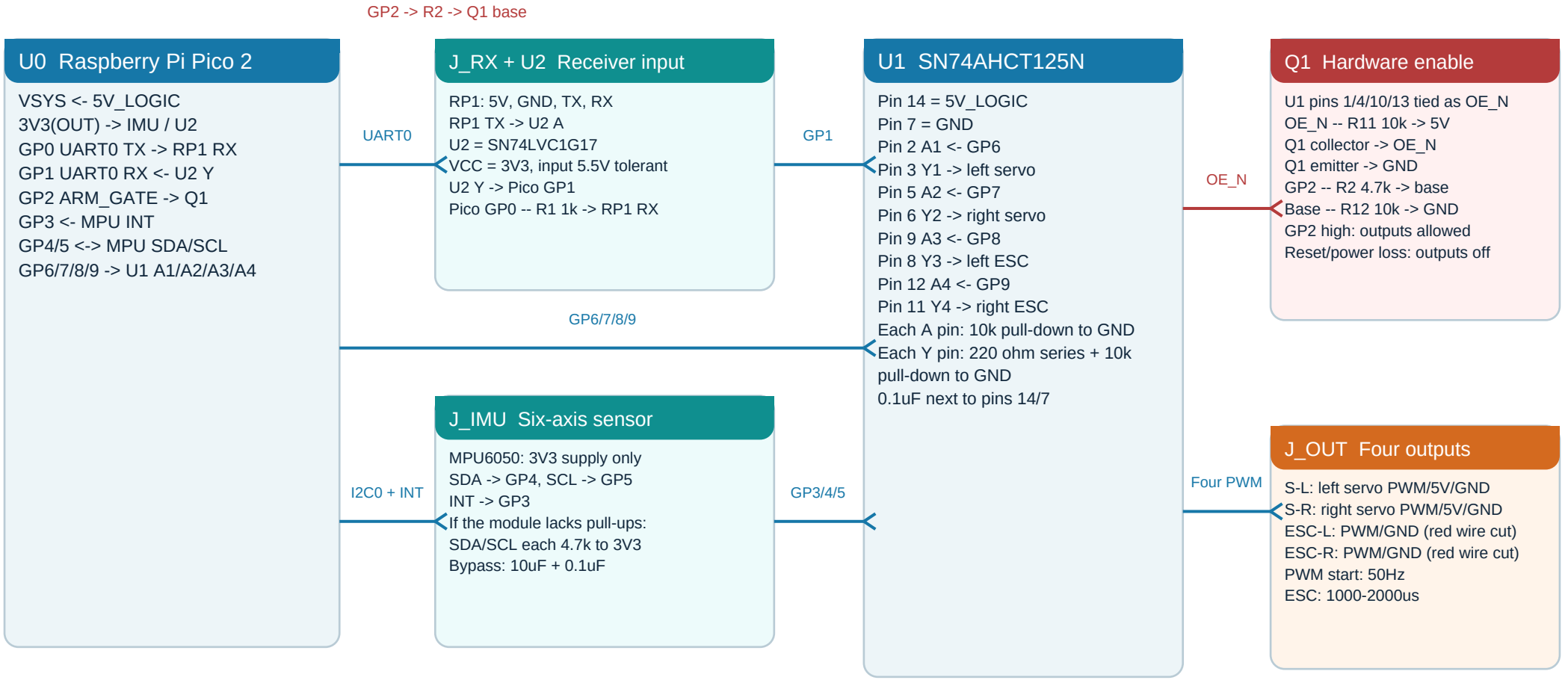
Key rules: power leads short and thick; motor phase wires may be extended. The 5V supply splits into two branches at the UBEC star point. All devices ultimately share ground, but motor current never flows through the Pico perfboard.

Capacitors and placement

C1 470uF/10V low-ESR: 5V servo star point; C2 470uF/10V low-ESR + C3 0.1uF: perfboard 5V entry; C4 10uF/25V + C5 0.1uF: next to the MPU6050. When ESC battery leads exceed 100mm, fit a low-ESR capacitor rated at least 16V at the ESC input.

Control board schematic

On any Pico fault, the four outputs are disabled automatically in hardware



Fail-safe logic: OE_N of U1 is pulled up to 5V by a 10k resistor; only when Pico GP2 is high and Q1 conducts is OE_N pulled low.
Therefore, whenever the Pico has not started, is reset, loses power, or the firmware deliberately drives GP2 low, U1 is high-impedance and the four outputs are pulled to GND by 10k resistors.

Harness, connectors and perfboard layout

Every PWM line is routed as a pair with a GND lead; power leads never enter the perfboard

Connector definitions

ID	Suggested connector	Connects to	Pins and notes
J1	XT30	BATT+ / BATT-	2S battery input, power star point
J2 / J3	2-pin power + 2-pin signal	ESC-L / ESC-R	Power wired straight to the PDB; signal is PWM+GND only
J4	2-pin	UBEC-IN	To the PDB, short input leads
J5	2-pin	UBEC-OUT	To 5V_STAR
J6	4-pin	RP1	5V / GND / TX / RX
J7	5-pin	MPU6050	3V3 / GND / SDA / SCL / INT
J8 / J9	3-pin servo	Left / right servo	SIG / 5V / GND
J10 / J11	3-pin shell, 2 pins used	Left / right ESC signal	SIG / unused / GND; red wire not connected

Perfboard physical layout

Left: 5V/GND input, C2 470uF, C3 0.1uF.
Center: Pico socket. Beside the Pico: U2 and the RP1 socket.
Right: U1, Q1, R2/R11/R12 and the four PWM output headers.
5V and GND each use one continuous tinned copper bus wire; the MPU6050 does not sit on the perfboard - mount it separately near the center of gravity.

Harness and interference control

Battery to ESC: short, thick, parallel; when relocation is needed, extend the three motor phase wires first.
PWM: twist each signal with its GND or run them side by side.
RP1 antenna: at least 30mm from the UBEC, ESCs, motors and carbon rods; keep the two antennas as close to 90 degrees as possible.
MPU6050: away from power leads and the UBEC; its axes must be parallel to the airframe axes.

Added small-parts BOM

Ref	Part	Qty	Purpose
U0	Raspberry Pi Pico 2	1	Headerless version saves mass; 5V into VSYS
U1	SN74AHCT125N DIP-14	1	5V PWM output buffer; second unit as spare
U2	SN74LVC1G17DBV + SOT23-to-DIP adapter	1	RP1 TX input protection, 3.3V supply
Q1	2N3904 or S8050 NPN	1	Hardware enable switch for U1
C1,C2	470uF / 10V low-ESR	2	Servo star point, logic board entry
C3,C5,C6,near U2	0.1uF ceramic	4	Close to the respective chip/module supply pins
C4	10uF / 25V X7R	1	Next to the MPU6050
R1	1k	1	Pico GP0 -> RP1 RX
R2,R17,R18	4.7k	3	Q1 base; use R17/R18 when I2C pull-ups are missing
R3-R6	220 ohm	4	Series resistors in the four PWM lines
R7-R16	10k	10	U1 input/output state and enable protection

First power-up acceptance sequence

1. Do not connect the Pico, receiver, IMU, servos or ESCs. Verify with a multimeter that 5V-GND, 3V3-GND and BATT+-BATT- show no shorts.
2. From a current-limited supply, feed the UBEC at 7.4V, 0.2A. Confirm the UBEC output is 4.75-5.25V; the Pico 3V3 rail is 3.20-3.35V.
3. Plug in the Pico without running output firmware: measure OE_N at about 5V and all four U1 Y pins at about 0V. Stop all further debugging if this step fails.
4. Connect the RP1 and MPU6050; confirm CRSF frames, I2C detection and IMU axis orientation. Then connect the servos one by one and check directions.
5. Connect both ESCs prop-less: calibrate minimum throttle; on radio lost link, receiver unplugged or Pico reset, OE_N must return to about 5V and the motors must stop.
6. Only when 5V stays >= 4.75V and the Pico does not reboot during full-travel dual-servo and low-throttle dual-ESC operation may props be fitted for the powered bench.